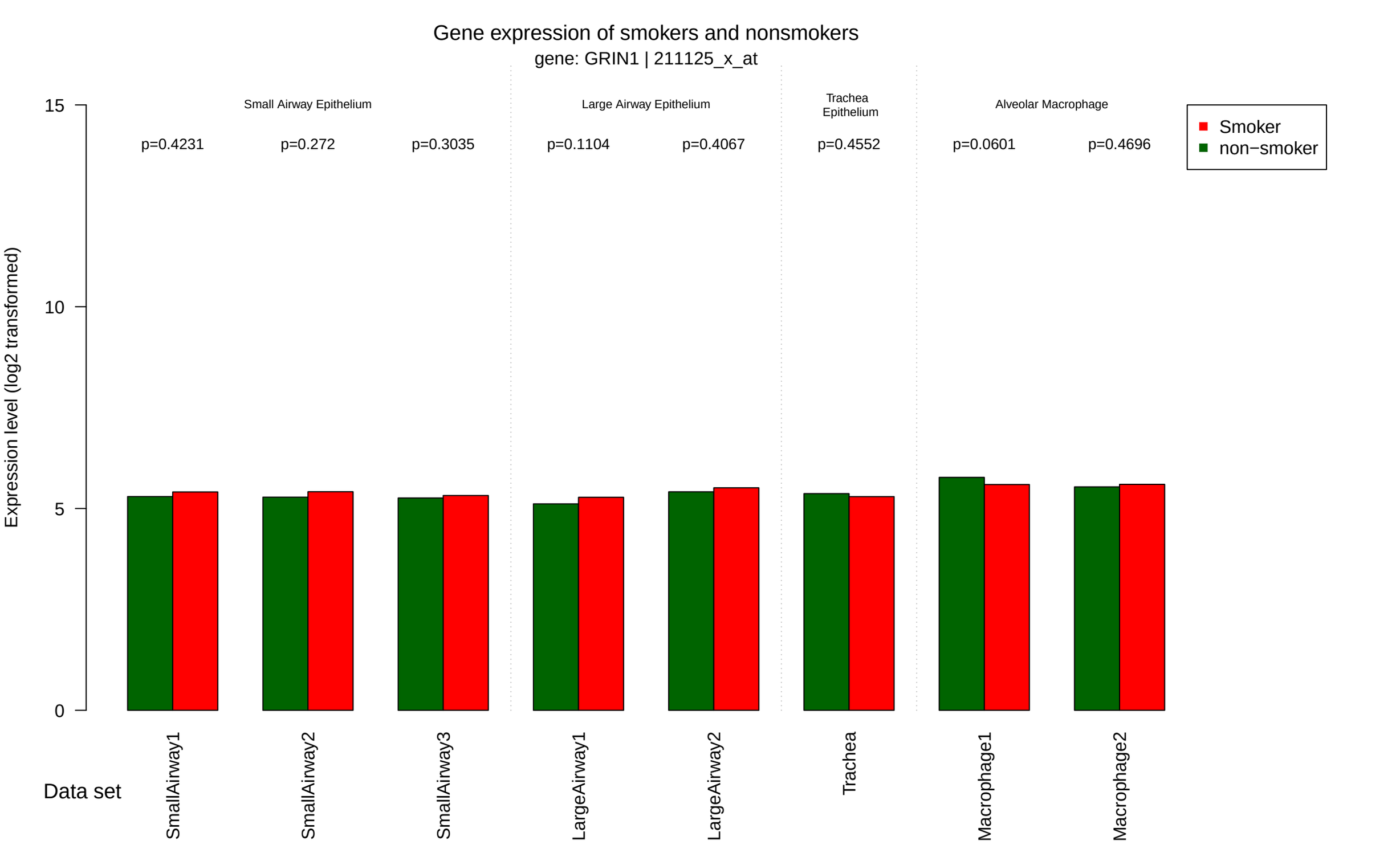
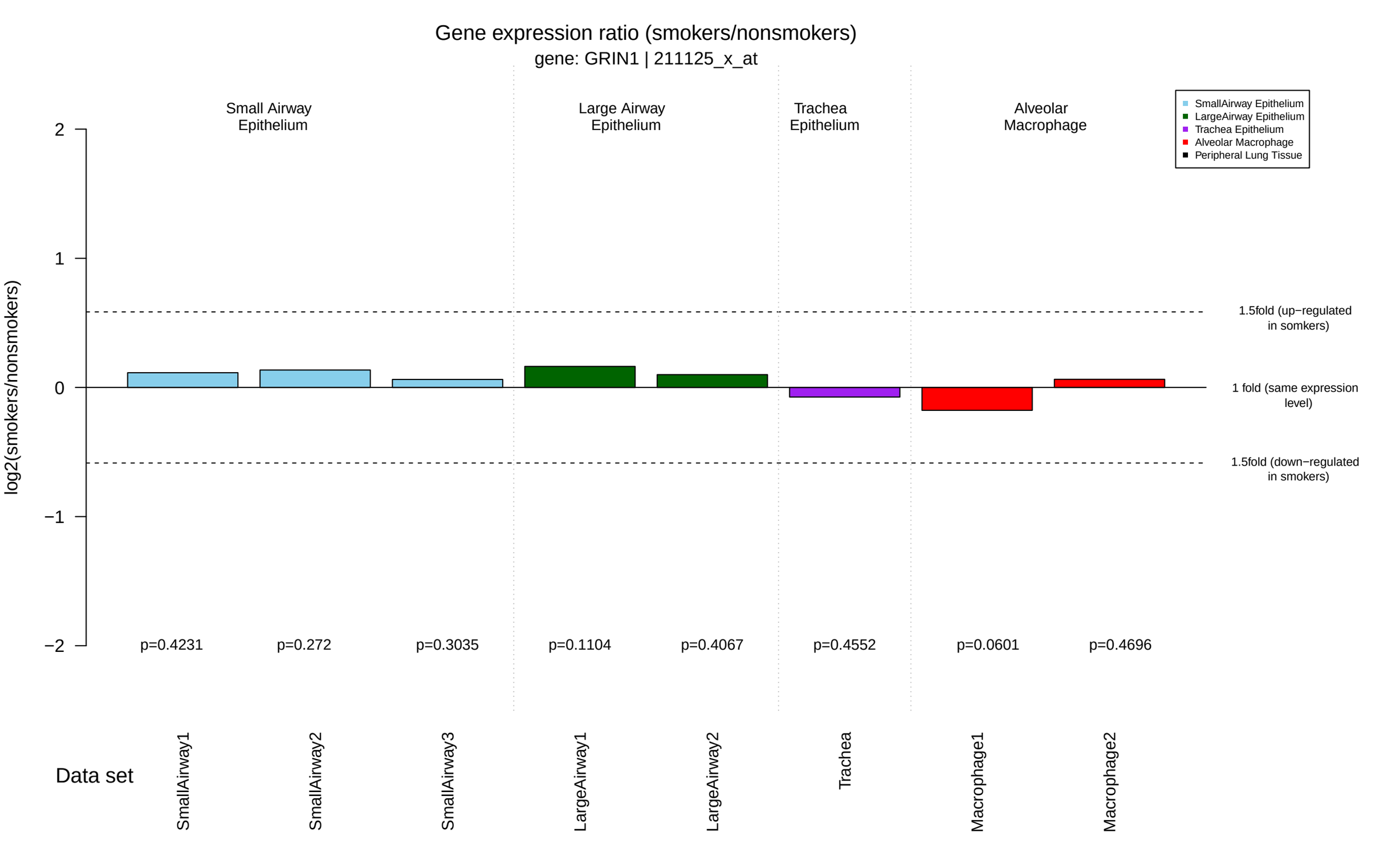
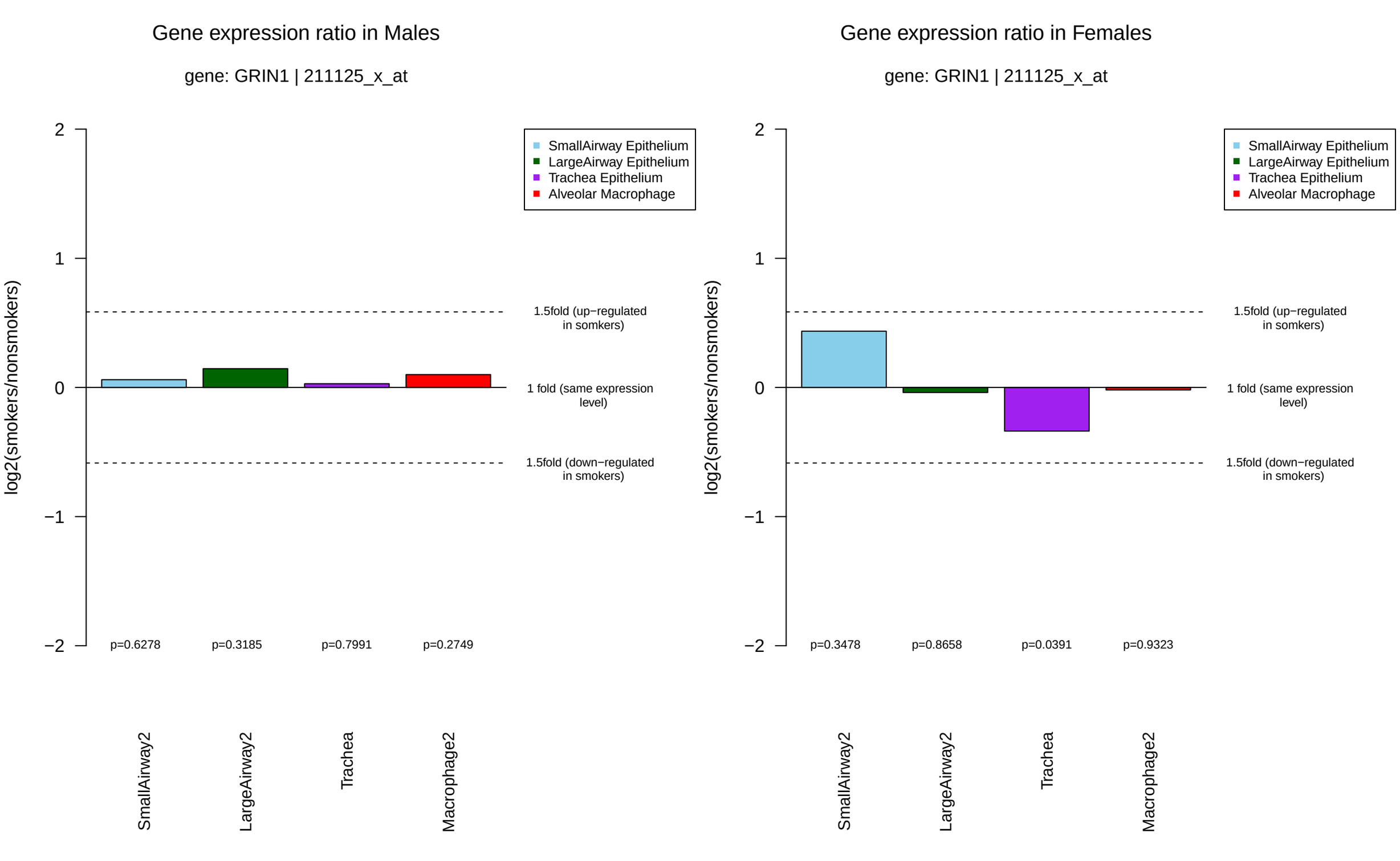
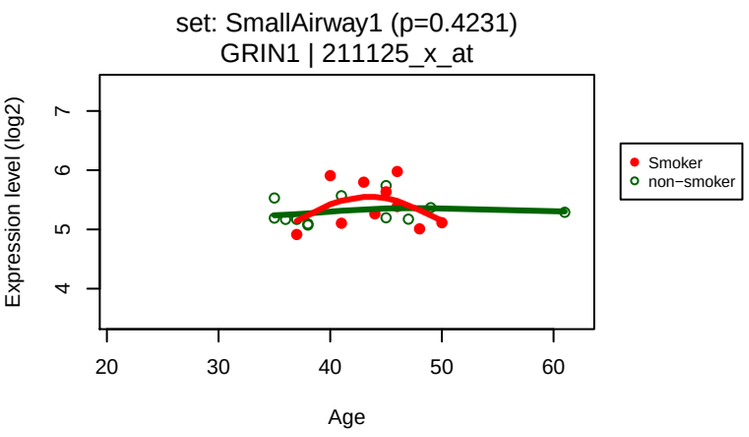
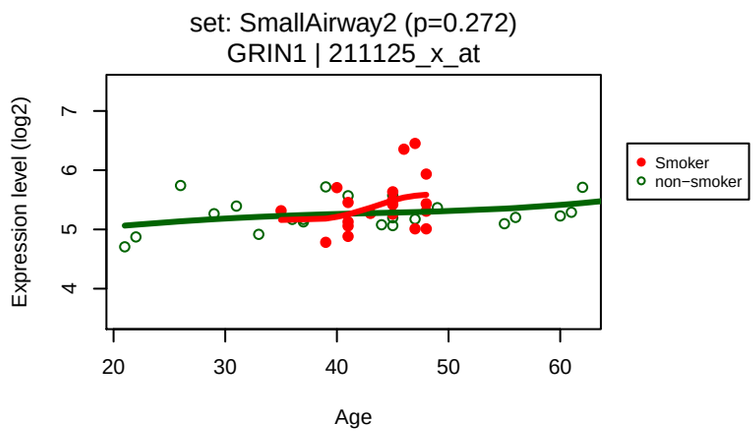
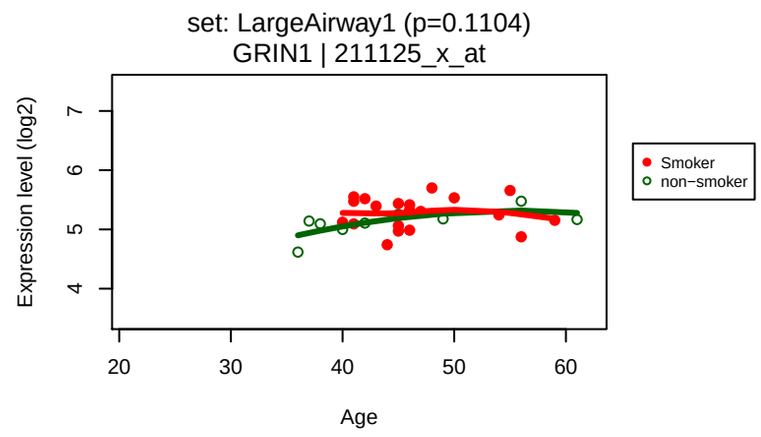

Gene expression of smokers and nonsmokers
gene: GRIN1 | 211125_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4231
p=0.272
p=0.3035
p=0.1104
p=0.4067
p=0.4552
p=0.0601
p=0.4696
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

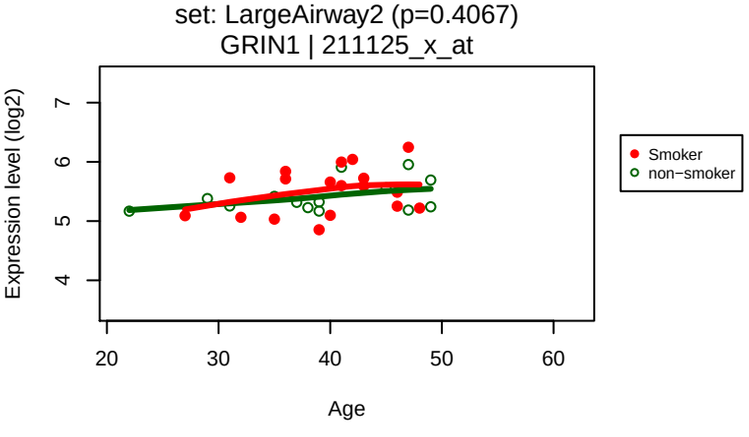

Gene expression ratio (smokers/nonsmokers)
gene: GRIN1 | 211125_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4231
p=0.272
p=0.3035
p=0.1104
p=0.4067
p=0.4552
p=0.0601
p=0.4696
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

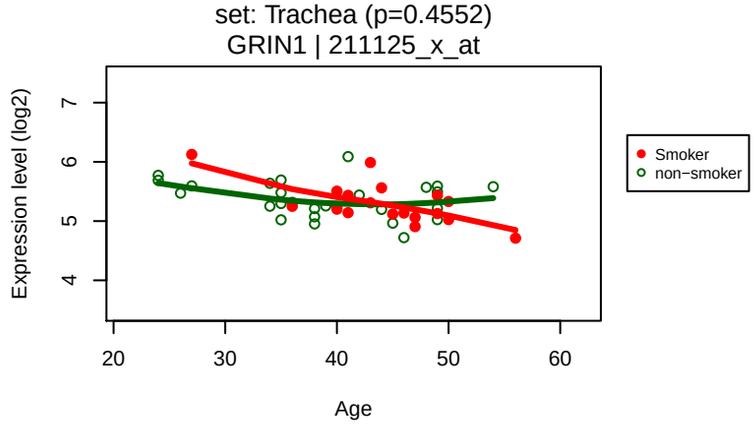

Gene expression ratio in Males
gene: GRIN1 | 211125_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6278
p=0.3185
p=0.7991
p=0.2749
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRIN1 | 211125_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3478
p=0.8658
p=0.0391
p=0.9323
SmallAirway2
LargeAirway2
Trachea
Macrophage2

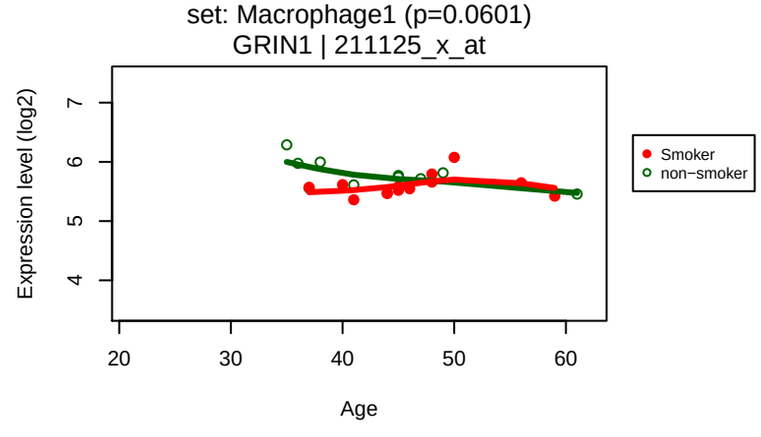

set: SmallAirway1 (p=0.4231)
GRIN1 | 211125_x_at
Expression level (log2)
Age
Smoker
non−smoker

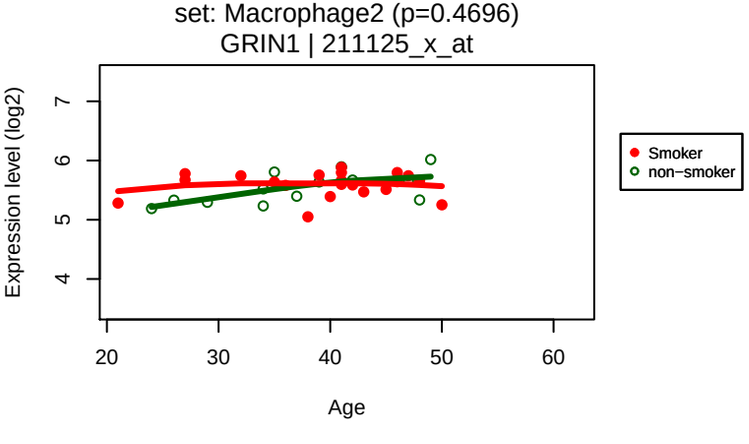

set: SmallAirway2 (p=0.272)
GRIN1 | 211125_x_at
Expression level (log2)
Age
Smoker
non−smoker

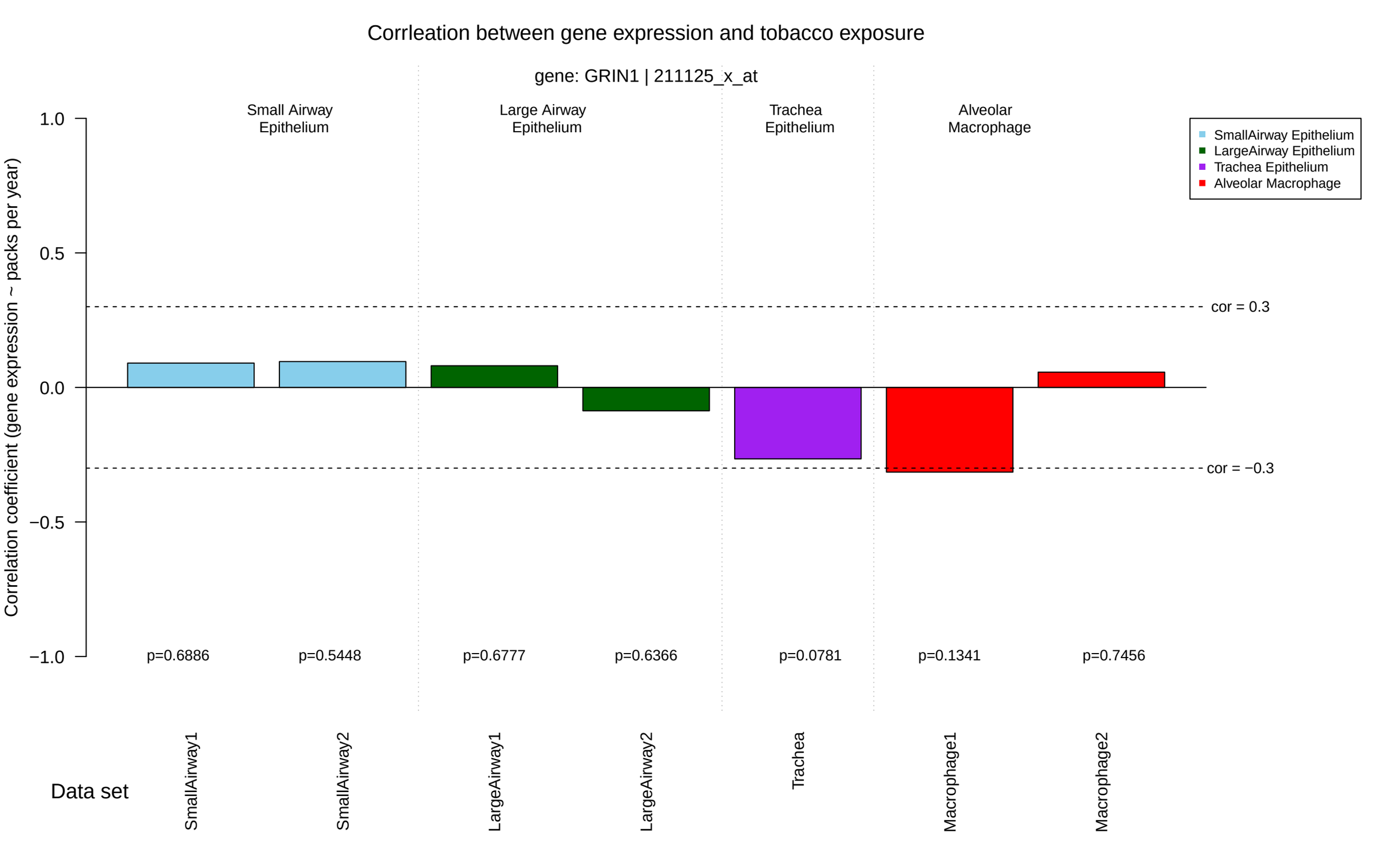

set: LargeAirway1 (p=0.1104)
GRIN1 | 211125_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4067)
GRIN1 | 211125_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4552)
GRIN1 | 211125_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0601)
GRIN1 | 211125_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4696)
GRIN1 | 211125_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRIN1 | 211125_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6886
p=0.5448
p=0.6777
p=0.6366
p=0.0781
p=0.1341
p=0.7456
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2